Code Coverage Report — Online Examination Portal

Course: UE24CS341A — Software Engineering, PES University, Dept. of CSE
Project: Online Examination System with Multiple-Choice Questions and Evaluation
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Build measured: commit `2d74453` · **Baseline:** `Documents/Test_Plan.pdf` v4.0

1. Summary

The automated test suite runs **88.8% of the application's statements** (1,101 of 1,240). One file accounts for over a third of the rest: a developer command that seeds sample questions, which no test runs and no user can reach. Leave it out and coverage is **92.7%** (1,101 of 1,188).

Two of the app modules are fully covered: `evaluation`, which does all the marking, and `integrity`. `results` is at 96.8%. What remains uncovered is mostly Administrator course editing and unenrolment, refusal messages for unusual requests, and pages reached only by a plain GET. Each uncovered block is listed in §5 with what it does.

Branch coverage, the stricter measure, has its own report: `Documents/Branch_Coverage_Report.pdf`.

2. How coverage was measured

Tool	Coverage.py 7.16.2 on Python 3.10, Django 5.2.18
Command	`coverage run manage.py test -v 2`, then `coverage report`
Configuration	`.coveragerc` at the repository root, committed with this report so the numbers can be reproduced
Tests run	227: all passed, none skipped (the same run that filled the Test Plan's results)
Measured	Every application module under the repository root
Not measured	Database migrations, the test files themselves, `manage.py`, the WSGI/ASGI entry points, and the `loadtest/` scripts (NFR-002 tooling, not application code)

A statement counts as covered if at least one test executed it. Coverage shows which code ran, not whether its result was checked. That check is the job of the assertions in each test, and those tests are keyed to Test Plan IDs.

3. Overall result

Measure	Covered	Total	Coverage
Statements, whole application	1,101	1,240	**88.8%**
Statements, without the seed command	1,101	1,188	**92.7%**
Combined statement + branch figure printed by `coverage report`	1,300	1,516	85.8%

4. Coverage by module

Each Django app maps to a part of the SRS, so the table shows which requirements areas the tests reach most thoroughly.

Module	What it holds	Statements covered	Coverage

Module	What it holds	Statements covered	Coverage
`evaluation`	Marking engine: FR-022, FR-023, FR-024	23 / 23	**100%**
`integrity`	Focus-loss events and audit-log viewer: FR-028, FR-029, FR-030	46 / 46	**100%**
`attempts`	Starting, answering, autosave, resume, auto-submission: FR-015 to FR-021	166 / 170	97.6%
`results`	Withholding and publishing results, scorecard: FR-025 to FR-027	60 / 62	96.8%
`portal`	Settings, URL routing, error pages, traceability matrix	45 / 47	95.7%
`questionbank`	Question authoring, tagging, versioning: FR-007 to FR-009	170 / 184	92.4%
`examsetup`	Building, publishing and scheduling examinations: FR-010 to FR-014	176 / 193	91.2%
`identity`	Registration, sign-in, roles, courses, enrolment: FR-001 to FR-006	235 / 273	86.1%
`repository`	Data model, audit writer, course form, seed command	180 / 242	74.4%
`repository` without the seed command		180 / 190	94.7%
Total		**1,101 / 1,240**	**88.8%**

5. What is not covered

These are the 139 uncovered statements, grouped by what the code does.

Where	Lines	What the code does	Assessment
`repository/management/commands/seed_bank.py`	11–198	Developer command that fills a demo question bank	No user can reach it. No test needed
`identity/views.py`	99–129	Administrator edits a course's code or title, with an audit entry	**Gap.** Real FR-004 behaviour that no test exercises
`identity/views.py`	244–263, 176	Faculty or Administrator unenrols a Student, including the "not your course" refusal	**Gap.** FR-005 behaviour. IT_011 now tests direct access, not unenrolment
`identity/views.py`	29–35	Database-level duplicate caught during registration	Runs only when two registrations race past form validation. IT_002 checks the outcome, but the form rejects the duplicate first
`examsetup/views.py`	28–33	The Faculty examination list page	**Gap.** Every test opens an examination directly
`examsetup/views.py`	45, 88, 139, 143, 180, 184	Refusals: an examination on an archived course, or changes to an examination no longer in Draft	Defensive paths, not exercised
`examsetup/views.py`	57, 110, 165, 225	Showing an empty form on a plain GET	Low risk. The POST paths around them are tested
`questionbank/views.py`	99, 114, 125, 127, 134, 140	The question API rejecting malformed JSON or a non-POST request	Input-hardening paths, not exercised
`questionbank/services.py`	88, 92	Refusing to delete a question in a published examination; restoring the previous version on delete	**Gap.** FR-009 rules. IT_014 tests the "answered question" refusal only
`questionbank/forms.py`	35, 41, 51	Rejecting zero or negative marks values	**Gap.** Field validation without a unit test
`repository/audit.py`	12, 17, 33	Two helper functions that nothing calls, and the "anonymous actor" normalisation	The helpers are unused code and can be deleted
`attempts/services.py`	127, 131	Dashboard grouping of an examination that has no window yet, or whose window ended without a submission	**Gap.** UT_038 covers "completed" only through a submitted attempt
`portal/error_handlers.py`	5, 17	The 400 and 500 error pages	Not triggered by any test
Remaining lines in `attempts/views.py`, `examsetup/forms.py`,	single lines	Model `__str__` methods, `create_superuser`, empty-form GETs, the duplicate-question check when	Low risk

Where	Lines	What the code does	Assessment
`identity/`, `repository/models.py`, `results/views.py`		assembling a paper, the post-sign-in `next` redirect, single-line refusal messages	

6. Conclusion

At 92.7% statement coverage without the seed command, every line of the marking and integrity code runs under test, and nearly every line of the attempt and results code. Those are the parts of the portal where a defect would cost a Student marks.

The **gaps** marked in §5 are real behaviour without a test: course editing, unenrolment, the published-question delete refusal, marks validation, the examination list page, and the dashboard's "missed examination" group. They are the follow-up list for the optional Sprint 3, and each would add a test keyed to a new or existing Test Plan ID.